



Modelling Financial Markets

Seminar 2: Market Efficiency — Hands-on



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIST: Borsa İstanbul (BIST 100 index) (the Istanbul Stock Exchange (BIST 100 index))
- ▣ BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CD: Chow–Denning (multiple variance-ratio test)
- ▣ CME: Chicago Mercantile Exchange
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DFA: Detrended Fluctuation Analysis
- ▣ FOMC: Federal Open Market Committee
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ FWER: Family-Wise Error Rate
- ▣ GMM: Generalized Method of Moments
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ IPC: Índice de Precios y Cotizaciones (the Mexican Stock Exchange index)
- ▣ MSPE: Mean Squared Prediction Error
- ▣ OLS: Ordinary Least Squares
- ▣ PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ RW1: Random Walk 1 (i.i.d. increments)
- ▣ RW3: Random Walk 3 (uncorrelated increments)
- ▣ SE: Standard Error
- ▣ TOM: Turn Of the Month
- ▣ US: United States

Acronyms used in this material (2/2)

- VR: Variance Ratio
- WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Outline

Introduction

Key Definitions

Part A: Pen and Paper

Part B: Computation and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive the null distribution of the variance ratio and the Stambaugh bias; compute runs tests and Hurst exponents step by step
- ▣ Test the random walk with inference robust to heteroskedasticity
- ▣ Measure time-varying efficiency with rolling windows and bootstrap bands
- ▣ Test calendar anomalies with HAC errors and multiple-testing corrections (Holm, StepM)
- ▣ Evaluate return predictability out of sample and event studies with clustered event days

Structure

- ▣ **Part A:** six pen-and-paper exercises and derivations (A1–A6)
- ▣ **Part B:** nine computational exercises with inference (B1–B9)
- ▣ **Part C:** one open question, a seed for the team project (C1), an AI answer to critique (C2) and a research case study project (C3)

- ▣ Background: Lecture 2; textbooks [Campbell, Lo & MacKinlay \(1997\)](#), Ch. 2 and [Franke, Härdle & Hafner \(2019\)](#)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions: ask what would change the conclusion
- Report every statistic with its uncertainty (standard error, band or p -value)
- Always compare a classical test with its robust version: when do they disagree, and why?

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (variance ratio, step by step), A3 (runs test, step by step), A5 (Hurst exponent from a variance ratio)
- Part B: B1 (homoskedastic vs robust VR), B3 (rolling DFA with a wild-bootstrap null), B5 (turn of the month with HAC errors)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A1, A5), A6 (model: A5)
- Part B: B2 (model: B1), B4 (model: B3, B1), B6 (model: B5), B7 (model: B1), B8 (model: B5, A4), B9 (model: Seminar 0 Ex. 7, B5)
- Part C: C1, open question and project seed; C2, critique an AI answer (model: A1, B1, B7); C3, the lecture case study at daily frequency (model: B5, Seminar 0 Ex. 7)

Setup: Python Environment and Data

-  **Open in Colab:** [chapter2_seminar_notebook.ipynb](#)
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`
- Series: S&P 500, BET, BET-TR, WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai, Euro Stoxx 50, Bitcoin; daily closes up to 18 September 2026
- Equity indices: weekdays only; a close equal to the previous one is treated as a filled holiday and removed (this also drops the rare genuine zero-return day); crypto keeps all 7 days of the week
- Returns: daily log returns $r_t = \ln P_t - \ln P_{t-1}$, each series on its own calendar

Key Definitions (1/3): Random Walk Tests

- **RW3**: uncorrelated increments, volatility may change; the hypothesis tested today
- **Variance ratio** Lo & MacKinlay (1988): $VR(q) = \text{Var}(r_t + \dots + r_{t-q+1}) / (q \text{Var}(r_t)) = 1 + 2 \sum_{k=1}^{q-1} (1 - k/q) \rho_k$
 - ▶ $z(q)$: assumes constant variance; $z^*(q)$: robust to heteroskedasticity (formulas in Lecture 2)
- **Chow–Denning** Chow & Denning (1993): $\max_i |z^*(q_i)|$ over m horizons; 5% critical value 2.49 for $m = 4$
- **Runs test** Wald & Wolfowitz (1940): number of runs of equal signs R , $\mathbb{E}[R] = 2n_1 n_2 / n + 1$, $\text{Var}[R] = 2n_1 n_2 (2n_1 n_2 - n) / (n^2 (n - 1))$
- **Hurst exponent** H : scaling of the range, $0.5 = \text{no long memory (short memory allowed)}$; DFA exponent Peng et al. (1994); self-similar case $VR(q) = q^{2H-1}$

Key Definitions (2/3): Inference Tools

- ▣ **Robust SE of $\hat{\rho}_k$:** $\sqrt{\hat{\tau}_k}$, $\hat{\tau}_k = \sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$, with $e_t = r_t - \bar{r}$
- ▣ **HAC** (Heteroskedasticity and Autocorrelation Consistent) errors [Newey & West \(1987\)](#): valid when residuals are heteroskedastic and autocorrelated
- ▣ **Wild bootstrap** [Kim \(2006\)](#): multiply centred returns by random signs ± 1 ; keeps the volatility path, destroys autocorrelation
- ▣ **Block bootstrap** [Künsch \(1989\)](#): resample blocks of consecutive days; keeps short-run dependence
- ▣ **FWER**: probability of at least one false rejection; **Bonferroni**: $p_i < \alpha/M$; **Holm**: sorted $p_{(j)} < \alpha/(M - j + 1)$, stop at the first failure

Key Definitions (3/3): Predictability and Event Studies

- **Stambaugh bias** Stambaugh (1999): $\mathbb{E}[\hat{\beta} - \beta] \approx -(\sigma_{uv}/\sigma_v^2)(1 + 3\rho)/T$ for a persistent predictor $x_{t+1} = \theta + \rho x_t + v_{t+1}$
- **Out-of-sample R^2** Welch & Goyal (2008): $R_{OS}^2 = 1 - \sum (r_t - \hat{r}_t)^2 / \sum (r_t - \bar{r}_t)^2$, $\bar{r}_t$ the recursive historical mean
- **Clark–West** Clark & West (2007): t -test on $f_t = (r_t - \bar{r}_t)^2 - [(r_t - \hat{r}_t)^2 - (\bar{r}_t - \hat{r}_t)^2]$ for nested models
- **Prediction-error variance** MacKinlay (1997): $\sigma_\varepsilon^2[1 + 1/T_0 + (r_{m,\tau} - \hat{\mu}_m)^2/((T_0 - 1)\hat{\sigma}_m^2)]$
- **StepM** Romano & Wolf (2005): step-down multiple testing with the joint bootstrap distribution of all statistics

A1: Variance Ratio, Step by Step [Solved]

Data and assumptions

- Eight daily returns in %, in time order: 1.0, −0.5, 0.8, 0.3, −1.2, 0.6, 0.9, −0.4; no data file is needed.
- Use the simple estimator without bias corrections: divide sums of squares by the number of terms.

Tasks

- (a) Compute the mean $\hat{\mu}$ and the one-day variance $\hat{\sigma}_1^2 = \frac{1}{T} \sum_t (r_t - \hat{\mu})^2$ with $T = 8$.
- (b) Form the seven overlapping two-day sums $s_t = r_t + r_{t-1}$ and compute $\hat{\sigma}_2^2 = \frac{1}{T} \sum_t (s_t - 2\hat{\mu})^2$.
- (c) Compute $\widehat{VR}(2) = \hat{\sigma}_2^2 / (2\hat{\sigma}_1^2)$, the lag-1 autocorrelation $\hat{\rho}_1$, and compare $\widehat{VR}(2)$ with $1 + \hat{\rho}_1$.

Report and interpret

- Report $\hat{\mu}$, $\hat{\sigma}_1^2$, $\hat{\sigma}_2^2$, $\widehat{VR}(2)$ and $\hat{\rho}_1$ with three decimals; say whether the series shows momentum or mean reversion, and why eight observations cannot support a test.

A1: Solution [Solved]

Step by step

- Mean: $\hat{\mu} = (1.0 - 0.5 + 0.8 + 0.3 - 1.2 + 0.6 + 0.9 - 0.4)/8 = 1.5/8 = 0.1875\%$.
- One-day variance: subtract $\hat{\mu}$ from each return, square, average over 8 terms: $\hat{\sigma}_1^2 = 0.5586$.
- Two-day sums: 0.5, 0.3, 1.1, -0.9, -0.6, 1.5, 0.5; subtract $2\hat{\mu} = 0.375$, square and average over 7 terms: $\hat{\sigma}_2^2 = 0.6292$.
- Variance ratio: $\widehat{VR}(2) = 0.6292/(2 \times 0.5586) = 0.563$.
- Autocorrelation: $\hat{\rho}_1 = \sum_t e_t e_{t-1} / \sum_t e_t^2 = -0.395$, so $1 + \hat{\rho}_1 = 0.605$; the gap to 0.563 comes from the end terms that a sample of eight cannot average out.

Interpretation

- $\widehat{VR}(2) < 1$ and $\hat{\rho}_1 < 0$: consecutive returns tend to reverse, i.e. mean reversion. With $T = 8$ the standard error of $\hat{\rho}_1$ is about $1/\sqrt{8} = 0.35$, so the value is not significant: the exercise trains the mechanics, not the inference.

A2: Derive the Variance Ratio and Its Null Distribution (1/2) [Proposed]

Solution discussed in class

Data and assumptions

- ▣ Returns r_t are covariance-stationary with variance γ_0 and autocorrelations ρ_k . Under RW1 (i.i.d.) the sample autocorrelations satisfy $\sqrt{T}(\hat{\rho}_1, \dots, \hat{\rho}_{q-1}) \rightarrow N(0, I)$; under RW3 with the conditions of [Lo & MacKinlay \(1988\)](#), including $\mathbb{E}[\varepsilon_t^2 \varepsilon_{t-j} \varepsilon_{t-k}] = 0$ for $j \neq k$, the covariance is $\text{diag}(\delta_k)$.
- ▣ Numerical check: S&P 500, daily log returns 2000–2026, $T = 6\,714$, $\widehat{\text{VR}}(2) = 0.901$ (Lecture 2).

Tasks

- ▣ (a) Expand $\text{Var}(\sum_{j=0}^{q-1} r_{t-j})$ in autocovariances and derive $\text{VR}(q) = 1 + 2 \sum_{k=1}^{q-1} (1 - k/q) \rho_k$.
- ▣ (b) Apply the delta method to show that under RW1 $\sqrt{T}(\widehat{\text{VR}}(q) - 1) \rightarrow N(0, \frac{2(2q-1)(q-1)}{3q})$, and evaluate the variance for $q = 2, 5, 20$.

A2: Derive the Variance Ratio and Its Null Distribution (2/2) [Proposed]

Solution discussed in class

Tasks

- (c) Replace I by $\text{diag}(\delta_k)$ and write the asymptotic variance $\theta(q)$ used by the robust $z^*(q)$.
- (d) Compute $z(2)$ for the S&P 500 from the numbers above and decide at $\alpha = 5\%$.

Report and interpret

- Report the derivation in three lines, the three variances and $z(2)$; explain why the same weights $1 - k/q$ appear in the robust version. Model: A1.

A3: Runs Test, Step by Step [Solved]

Data and assumptions

- Signs of 20 consecutive daily returns: ++-+++--+-+---+---; zero returns do not occur. Under H_0 the signs are independent with a constant probability of a positive day [Wald & Wolfowitz \(1940\)](#).

Tasks

- (a) Count the positive signs n_1 , the negative signs n_2 and the runs R (maximal blocks of equal signs).
- (b) Compute $\mathbb{E}[R] = 2n_1n_2/n + 1$ and $\text{Var}[R] = 2n_1n_2(2n_1n_2 - n)/(n^2(n - 1))$ with $n = n_1 + n_2$.
- (c) Compute $z = (R - \mathbb{E}[R])/\sqrt{\text{Var}[R]}$, the two-sided p -value from the Normal distribution, and decide at $\alpha = 5\%$.

Report and interpret

- Report n_1 , n_2 , R , $\mathbb{E}[R]$, $\text{Var}[R]$, z and p ; say what too few or too many runs would mean for the dependence of signs.

A3: Solution [Solved]

Step by step

- (a) Counting: $n_1 = 12$ plus signs, $n_2 = 8$ minus signs, $n = 20$; the sign changes 10 times, so $R = 11$ runs.
- (b) $\mathbb{E}[R] = 2 \cdot 12 \cdot 8/20 + 1 = 10.60$; $2n_1n_2 = 192$ and $\text{Var}[R] = 192 \cdot (192 - 20)/(400 \cdot 19) = 4.345$, so $\text{SD}[R] = 2.085$.
- (c) $z = (11 - 10.60)/2.085 = 0.40/2.085 = 0.19$; two-sided $p = 0.85$.

Interpretation

- $p = 0.85 > 0.05$: do not reject independence; the observed number of runs is what chance produces.
- Too few runs ($z < 0$) would mean persistent signs (positive dependence of signs, not necessarily of returns); too many ($z > 0$), alternating signs (sign reversal). With $n = 20$ the Normal approximation is rough: exact tables are preferable.

A4: The Stambaugh Bias [Proposed]

Solution discussed in class

Data and assumptions

- ▣ Predictive system: $r_{t+1} = \alpha + \beta x_t + u_{t+1}$ and $x_{t+1} = \theta + \rho x_t + v_{t+1}$; write $u = \phi v + e$ with $\phi = \sigma_{uv}/\sigma_v^2$ and $\mathbb{E}[e | x] = 0$; Kendall: $\mathbb{E}[\hat{\rho} - \rho] \approx -(1 + 3\rho)/T$ **Stambaugh (1999)**.
- ▣ Illustrative annual inputs: $\rho = 0.98$, $T = 60$ years, $\text{corr}(u, v) = -0.9$, $\sigma_u = 0.17$, $\sigma_v = 0.14$.

Tasks

- ▣ (a) Show that $\hat{\beta} - \beta = \phi(\hat{\rho} - \rho) + \sum \tilde{x}_t e_{t+1} / \sum \tilde{x}_t^2$, where $\tilde{x}$ is the demeaned predictor, and derive $\mathbb{E}[\hat{\beta} - \beta]$.
- ▣ (b) Compute ϕ , $\mathbb{E}[\hat{\rho} - \rho]$ and the bias of $\hat{\beta}$ with the inputs above.
- ▣ (c) Compute $\text{SE}(\hat{\beta}) \approx \sigma_u / (\sigma_x \sqrt{T})$ with $\sigma_x = \sigma_v / \sqrt{1 - \rho^2}$ and express the bias in standard errors.

Report and interpret

- ▣ Report ϕ , the bias, the SE and their ratio; explain why a persistent predictor with correlated innovations makes “significant” predictability too easy. Model: A1, A5.

A5: Hurst Exponent from a Variance Ratio [Solved]

Data and assumptions

- For exactly self-similar increments (fractional Gaussian noise) $VR(q) = q^{2H-1}$. Data: BET, daily log returns 2000–2026 (data/market/BET.csv, close), $\widehat{VR}(20) = 1.377$ (Lecture 2).

Tasks

- (a) Take logarithms of $VR(q) = q^{2H-1}$ and solve for H .
- (b) Compute $\hat{H}$ for BET with $q = 20$.
- (c) Compare it with the Anis–Lloyd corrected R/S estimate 0.533 of Lecture 2.

Report and interpret

- Report the formula and $\hat{H}$ with three decimals; say whether this value proves long memory.

A5: Solution [Solved]

Step by step

- (a) $\ln VR(q) = (2H - 1) \ln q$, hence $H = 0.5 + \ln VR(q)/(2 \ln q)$.
- (b) $\ln 1.377 = 0.3199$ and $2 \ln 20 = 5.991$, so $\hat{H} = 0.5 + 0.3199/5.991 = 0.553$.
- (c) The corrected R/S gives 0.533: both estimates lie above 0.5 and point to mild persistence.

Interpretation

- The formula assumes exact self-similarity. An AR(1) with $\rho_1 = 0.109$ (short memory) also gives $VR(20) > 1$ and hence $\hat{H} > 0.5$: this value alone does not prove long memory (see A6).

A6: Hurst Exponent of the S&P 500 [Proposed]

Solution discussed in class

Data and assumptions

- S&P 500, daily log returns 2000–2026 (data/market/GSPC.INDX.csv, close), $\widehat{VR}(20) = 0.753$; a short-memory benchmark: AR(1) with $\rho_1 = -0.1$, so $\rho_k = (-0.1)^k$.

Tasks

- (a) Compute $\hat{H}$ with the formula of A5 and say whether the index is persistent or anti-persistent.
- (b) For the AR(1), compute $VR(20) = 1 + 2 \sum_{k=1}^{19} (1 - k/20) \rho_k$ and the implied $H(20)$.
- (c) Show that for any short-memory process ($\sum_k |\rho_k| < \infty$) with $f(0) > 0$, $VR(q) \rightarrow 1 + 2 \sum_{k \geq 1} \rho_k = 2\pi f(0)/\gamma_0$, hence $H(q) \rightarrow 0.5$; check with $q = 250$.

Report and interpret

- Report $\hat{H}$, $VR(20)$, $H(20)$, $VR(250)$, $H(250)$; conclude whether $H < 0.5$ needs long memory. Model: A5.

B1: Homoskedastic vs Robust Variance Ratio [Solved]

Data and assumptions

- S&P 500 (GSPC.INDX) and BET (BET), daily closes 2000–2026 from data/market; weekdays only, unchanged closes treated as filled holidays and removed; daily log returns, $T = 6\,714$ and $6\,670$.
- H_0 for $z(q)$: RW1 (i.i.d.); H_0 for $z^*(q)$: RW3 (uncorrelated, heteroskedasticity allowed) Lo & MacKinlay (1988).

Tasks

- (a) Compute $\widehat{VR}(q)$ with the Lo–MacKinlay bias corrections for $q = 2, 5, 20$.
- (b) Compute the homoskedastic $z(q)$ and the robust $z^*(q)$ with their two-sided p -values and decide at $\alpha = 5\%$.
- (c) Run the Chow–Denning joint test over $q = 2, 5, 10, 20$ (critical value 2.49 at 5%) Chow & Denning (1993).

Report and interpret

- Report a table of $\widehat{VR}$, z , z^* and the CD statistic with its p -value; explain why $|z| > |z^*|$, whether the sign of $VR - 1$ agrees with $\hat{\rho}_1$, and whether you would trade the S&P 500 reversal.

B1: Solution (1/2) [Solved]

Step by step

	$\widehat{VR}(2)$	$z(2)$	$z^*(2)$	$\widehat{VR}(5)$	$z(5)$	$z^*(5)$	$\widehat{VR}(20)$	$z^*(20)$
S&P 500	0.901	-8.12	-3.64	0.830	-6.37	-2.77	0.753	-1.86
BET	1.108	8.83	3.84	1.173	6.44	2.98	1.377	3.31

- (a)–(b) Steps: overlapping q -day sums, $\hat{\sigma}_c^2(q)/\hat{\sigma}_a^2$, then $z = (\widehat{VR} - 1)/\sqrt{2(2q - 1)(q - 1)/(3qT)}$ and $z^* = \sqrt{T}(\widehat{VR} - 1)/\sqrt{\hat{\theta}(q)}$.
- S&P 500: $z^*(2) = -3.64$ ($p = 0.0003$), $z^*(5) = -2.77$ ($p = 0.0056$): reject RW3; $z^*(20) = -1.86$ ($p = 0.063$): do not reject.
- BET: $z^*(2) = 3.84$ ($p = 0.0001$), $z^*(20) = 3.31$ ($p = 0.0009$): reject at every horizon.

B1: Solution (2/2) [Solved]

Step by step

- (c) Chow–Denning = $\max_q |z^*(q)|$: S&P 500 3.64 ($p = 0.0011$), BET 3.84 ($p = 0.0005$); both exceed 2.49, so both reject RW3 jointly at 5%.

Interpretation

- $|z|$ is two to three times $|z^*|$: volatility clustering makes $\text{Var}(\hat{\rho}_k) > 1/T$, so the homoskedastic test overstates the evidence.
- Signs agree with $\hat{\rho}_1$: VR < 1 (reversal) for the S&P 500, VR > 1 (persistence) for BET, growing with q .
- A statistical rejection is not a trading rule: trading costs, risk and stability over time (B3, B7) are missing from the test.

B2: BET-TR and Emerging Markets [Proposed]

Solution discussed in class

Data and assumptions

- BET-TR (BETTR.INDX, 2014–2026) and the indices WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai (2000–2026), daily closes from data/market, each on its own calendar; daily log returns.

Tasks

- (a) For each series compute $\widehat{VR}(2)$, $z(2)$, $z^*(2)$ and $z^*(5)$ as in B1.
- (b) Compute the Chow–Denning statistic over $q = 2, 5, 10, 20$ and its p -value.
- (c) Count the rejections at $\alpha = 5\%$ with $z(2)$ and with Chow–Denning.

Report and interpret

- Report one table (one row per market); explain the gap between the two counts, why BET-TR can differ from BET (compare BET on BET-TR's period to separate dividends from sample), and whether emerging markets are less efficient as a group. Model: B1.

B3: Rolling DFA on the S&P 500 (1/2) [Solved]

Data and assumptions

- S&P 500 daily log returns 2000–2026 (GSPC.INDX); rolling windows of 500 trading days moved by 21 days; DFA exponent α_{DFA} (linear detrending) with box sizes from 10 to $T/4$ [Peng et al. \(1994\)](#).
- H_0 : RW3, i.e. no serial correlation, volatility clustering allowed.

Tasks

- (a) Compute α_{DFA} in each window.
- (b) In each window build the null band by wild bootstrap [Kim \(2006\)](#): 199 series $e_t \eta_t$, $\eta_t = \pm 1$ with probability 1/2, e_t the demeaned returns; keep the 2.5% and 97.5% quantiles.

B3: Rolling DFA on the S&P 500 (2/2) [Solved]

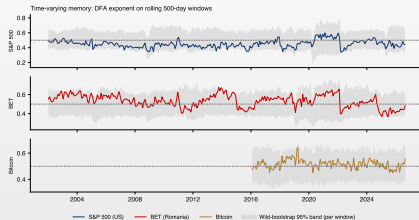
Tasks

- (c) Compute the share of windows outside their band and compare it with one fixed i.i.d. Student- t_4 band (300 simulations of length 500).

Report and interpret

- Report the chart, both shares, the mean, minimum and maximum exponent with dates; explain why the null must keep the volatility path and why the band is wider than the full-sample one.

B3: Solution (1/2) [Solved]



Step by step

- (a)–(b) 296 windows; the per-window band averages $[0.359, 0.644]$ and its width ranges from 0.208 to 0.500 (widest in the window ending 2022-02-11).

B3: Solution (2/2) [Solved]

Step by step

- (c) Outside their own band: 2.0% of windows, below the 5% expected by chance; with the fixed Student- t_4 band $[0.378, 0.614]$: 6.4%.
- Mean exponent 0.450; minimum 0.341 (window ending 2022-03-15); maximum 0.598 (2021-02-12); both extremes are in windows that contain the 2020 crash.

Interpretation

- No evidence of memory beyond chance: the mean below 0.5 matches the negative $\hat{\rho}_1$ (mild reversal). RW3 allows volatility clustering, which raises α_{DFA} ; sign flips keep $|e_t|$ and remove serial correlation, so the band adapts to each window.
- 500 observations instead of 6 714: the sampling error of a scaling exponent grows roughly like $1/\sqrt{T}$, hence the wide band.

B4: Adaptive Efficiency of Bitcoin [Proposed]

Solution discussed in class

Data and assumptions

- Bitcoin (BTC-USD.CC, close), daily log returns from 2011, 7 days a week; rolling windows of 365 days moved by 30 days; H_0 : RW3 Lo (2004).

Tasks

- (a) Compute the robust $z^*(5)$ of the Lo–MacKinlay test in each window.
- (b) In each window build a wild-bootstrap band for $z^*(5)$ Kim (2006): 99 replications of $e_t \eta_t$, $\eta_t = \pm 1$; keep the 2.5% and 97.5% quantiles.
- (c) Count the windows outside their band and outside ± 1.96 ; compare with the chance level $0.05 \times (\text{number of windows})$ and with Urquhart (2016).

Report and interpret

- Report the chart, the rejection dates with z^* and band, and both shares; say whether Bitcoin becomes more efficient over time and why the band is not ± 1.96 . Model: B3, B1.

B5: The Turn-of-the-Month Effect with HAC Errors [Solved]

Data and assumptions

- S&P 500 and BET, daily log returns 2000–2026 in basis points (bp); $\text{TOM}_t = 1$ on the last trading day of a month and the first three trading days of the next [Ariel \(1987\)](#), 0 otherwise.

Tasks

- (a) Estimate $r_t = a + b \text{TOM}_t + u_t$ by OLS; b is the extra return on TOM days.
- (b) Compute the OLS and the Newey–West HAC standard errors of $\hat{b}$ with 5 lags [Newey & West \(1987\)](#), the two t -statistics and the HAC p -value; decide at $\alpha = 5\%$.
- (c) Convert $\hat{b}$ into an extra return per month (4 TOM days).

Report and interpret

- Report a table (share of TOM days, $\hat{a}$, $\hat{b}$, both SEs and t); explain why OLS and HAC errors differ and which explanations (flows, pension funds, rebalancing) you would test next.

B5: Solution (1/2) [Solved]

Result

	TOM share	$\hat{a}$ (bp)	$\hat{b}$ (bp)	SE_{OLS}	t_{OLS}	SE_{HAC}	t_{HAC} (p)
S&P 500	19.1%	1.9	3.2	3.8	0.86	3.5	0.93 (0.35)
BET	19.2%	3.1	17.3	4.3	3.99	4.8	3.62 (0.0003)

Step by step

- (a) $\hat{a}$ is the mean on other days, $\hat{a} + \hat{b}$ the mean on TOM days: S&P 500 5.1 vs 1.9 bp; BET 20.4 vs 3.1 bp.
- (b) BET: $t_{HAC} = 17.3/4.8 = 3.62$, $p = 0.0003$: reject $b = 0$. S&P 500: $t_{HAC} = 0.93$, $p = 0.35$: do not reject.

B5: Solution (2/2) [Solved]

Step by step

- (c) BET: $4 \times 17.3 \approx 69$ bp extra per month; S&P 500: $4 \times 3.2 \approx 13$ bp, not distinguishable from zero.

Interpretation

- For BET the HAC error exceeds the OLS error (volatility clustering and $\rho_1 > 0$), so t falls from 3.99 to 3.62; the effect survives and is economically large.
- One market and one effect were chosen among many: before believing it, correct for the number of effects tried (B6), then look for a mechanism such as salary and pension inflows at the start of the month.

B6: Fifteen Calendar Tests and Multiple Testing (1/2) [Proposed]

Solution discussed in class

Data and assumptions

- Five markets (S&P 500, BET, Euro Stoxx 50, WIG20, BUX), daily log returns 2000–2026, each on its own calendar; three effects: day of the week (Wald test of equal means), January (monthly returns, HAC with 3 lags), turn of the month (daily, HAC with 5 lags).

Tasks

- (a) Compute the 15 p -values and the raw decisions at $\alpha = 5\%$.
- (b) Apply Bonferroni ($p_i < \alpha/15$) and Holm (sorted $p_{(j)} < \alpha/(16 - j)$), stop at the first failure).

B6: Fifteen Calendar Tests and Multiple Testing (2/2) [Proposed]

Solution discussed in class

Tasks

- (c) Apply Romano–Wolf StepM [Romano & Wolf \(2005\)](#): resample blocks of 3 calendar months, the same for all markets, 999 times; in each replication re-estimate coefficients and HAC errors and compute p -values of statistics centred at the original estimates; critical value: the 5% quantile of the minimum bootstrap p -value over the hypotheses still active; reject those below it and repeat until no new rejection.

Report and interpret

- Report the table of p -values with the four decisions (raw, Bonferroni, Holm, StepM), the FWER of 15 independent tests, and which effects pass $|t| > 3$ [Harvey, Liu & Zhu \(2016\)](#); say whether StepM buys power over Holm here, and why. Model: B5.

B7: Is the S&P 500 Reversal a Crisis Effect? [Proposed]

Solution discussed in class

Data and assumptions

- S&P 500 daily log returns 2000–2026 ($T = 6\,714$); crisis days: 1 Sep 2008 – 30 Jun 2009 and 15 Feb – 30 Jun 2020.

Tasks

- (a) Compute $\hat{\rho}_1$ with the i.i.d. $t = \hat{\rho}_1 \sqrt{T}$ and the robust $t = \hat{\rho}_1 / \sqrt{\hat{\tau}_1}$ on all days.
- (b) Remove the crisis days and repeat (a) within the contiguous retained segments: no lag pair and no 5-day sum may span a removed period; report the share of days removed.
- (c) Compute $z^*(5)$ in both samples and decide at $\alpha = 5\%$.

Report and interpret

- Report a two-row table (T , $\hat{\rho}_1$, both t , $z^*(5)$); say how much of the autocorrelation comes from the crises, why the i.i.d.–robust gap shrinks, and when removing crisis days becomes data snooping. Model: B1.

B8: Predictive Regression and an Out-of-Sample Test (1/2) [Proposed]

Solution discussed in class

Data and assumptions

- S&P 500 monthly log excess return (dividends included, minus the 3-month T-bill rate, FRED TB3MS), 1934–2026, on the lagged log dividend–price ratio $dp_t = \ln D_t - \ln P_t$ [Stambaugh \(1999\)](#).
- Prices and dividends: Shiller's monthly S&P Composite (to 2026-06), extended with S&P 500 closes (GSPC.INDX) and SPY dividends (SPY.US).

Tasks

- (a) Estimate $r_{t+1} = \alpha + \beta dp_t + u_{t+1}$ by OLS with Newey–West errors (12 lags); estimate the AR(1) of dp and $\text{corr}(\hat{u}, \hat{v})$; correct $\hat{\beta}$ for the Stambaugh bias (A4).
- (b) From January 1970 forecast recursively (expanding window) and compute R_{OS}^2 against the historical mean [Welch & Goyal \(2008\)](#).

B8: Predictive Regression and an Out-of-Sample Test (2/2) [Proposed]

Solution discussed in class

Tasks

- (c) Compute the Clark–West t (HAC, 12 lags, one-sided) Clark & West (2007) and apply the Campbell–Thompson restrictions Campbell & Thompson (2008).

Report and interpret

- Report $\hat{\beta}$, t , $\hat{\rho}$, the bias, R_{OS}^2 and the Clark–West t ; say whether the in-sample evidence survives, why Diebold–Mariano is wrong for nested models, and what $R^2 = 0.5\%$ is worth to a mean–variance investor. Model: B5, A4.

B9: The Bank Tax, TLV and BRD Jointly (1/2) [Proposed]

Solution discussed in class

Data and assumptions

- Banca Transilvania (TLV.R0) and BRD (BRD.R0), `adjusted_close`, and the BET index (BET, `close`), joined on common trading days; event: the bank tax announced on the evening of 18 December 2018, day $0 = 19$ December (Seminar 0, Ex. 7).
- Market model $r_{i,t} = a_i + b_i r_{m,t} + \varepsilon_{i,t}$ estimated on days $[-250, -11]$ ($T_0 = 240$).

Tasks

- (a) For each bank compute AR_0 , the naive $t = AR_0 / \hat{\sigma}_\varepsilon$ and the t with the prediction-error variance $\hat{\sigma}_\varepsilon^2 [1 + 1/T_0 + (r_{m,0} - \hat{\mu}_m)^2 / ((T_0 - 1)\hat{\sigma}_m^2)]$ MacKinlay (1997).
- (b) For the mean AR_0 compute the t assuming independent banks and the t of the equal-weighted portfolio; compute the residual correlation $\bar{r}$ and the Kolari–Pynnönen factor $\sqrt{(1 - \bar{r}) / (1 + (N - 1)\bar{r})}$ Kolari & Pynnönen (2010).

B9: The Bank Tax, TLV and BRD Jointly (2/2) [Proposed]

Solution discussed in class

Tasks

- (c) Compute $CAR[0, +5]$ and its t with the full covariance of the prediction errors.

Report and interpret

- Report one table (TLV, BRD, portfolio); say how much of “ -8.3% , $t = -10$ ” survives, why a common event day calls for the portfolio or Kolari–Pynnönen test, and whether the reversal contradicts semi-strong efficiency. Model: Seminar 0 Ex. 7, B5.

C1: Did the BVB Become More Efficient after the 2020 FTSE Upgrade? (1/2) [Proposed]

Solution discussed in class

Data and assumptions

- FTSE Russell (2019): Romania moved from Frontier to Secondary Emerging on 21 September 2020; more foreign investors and analysts should speed up price discovery. Background: Pele & Mazurencu-Marinescu (2012); the question extends Seminar 1, C1.
- Data: BET (data/market/BET.csv, close), daily log returns five years before and five years after the event; WIG20 and BUX as a comparison group.

Tasks

- (a) Compute $\hat{\rho}_1$ and $z^*(5)$ in both five-year windows.
- (b) Build a moving-block bootstrap 95% interval for the change in $\hat{\rho}_1$ (blocks of 20 days, 999 replications) Künsch (1989).

C1: Did the BVB Become More Efficient after the 2020 FTSE Upgrade? (2/2) [Proposed]

Solution discussed in class

Tasks

- (c) Add a rolling view (B3) and the same statistics for WIG20 and BUX to separate the event from common shocks; list the confounders (COVID-19, Hidroelectrica listed in 2023, liquidity).

Report and interpret

- Deliver a Quantlet folder in the team repository and one slide with the conclusion, the main limitation and the link with the Adaptive Markets Hypothesis [Lo \(2004\)](#).

C2: Critique an AI Answer (1/2) [Proposed]

Solution discussed in class

- **The prompt** a student sent to an AI (Artificial Intelligence) assistant: “Test whether the S&P 500 (daily closes, 2000–2026) follows a random walk with the variance ratio. Give Python code and interpret the result”
- **The answer:** the code below, followed by the interpretation

```
r = np.log(sp).diff().dropna()           # daily log returns
T = len(r)
def vr(q):
    rq = r.rolling(q).sum().dropna()      # overlapping q-day returns
    return rq.var() / (q * r.var())
q = 5
z = (vr(q) - 1) / np.sqrt(2 * (2*q - 1) * (q - 1) / (3 * q * T))
p = 2 * stats.norm.sf(abs(z))
```

- **Interpretation of the AI**
 - ▶ “VR(2) = 0.901, VR(5) = 0.830, $z(5) = -6.37$, $p < 0.001$: the random walk is rejected”
 - ▶ “Since VR(2) = $1 + 2\rho_1$, the first-order autocorrelation is $\rho_1 \approx -0.050$ ”
 - ▶ “VR < 1 means positive autocorrelation, i.e. momentum: buy the S&P 500 after up days”

C2: Critique an AI Answer (2/2) [Proposed]

Solution discussed in class

Task

- ▣ (a) Identify the conceptual errors and check whether the code implements the estimator specified in the lecture
- ▣ (b) For each error, say how you would detect it: a test, a published number, the lecture or the library documentation
- ▣ (c) Give the corrected code and the corrected interpretation
- ▣ (d) Log each error as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version
- ▣ Model: A1, B1 and B7

Why this exercise

- ▣ AI code usually runs without an error message: the mistakes are in the statistics, not in the syntax
- ▣ Course rule: you may use AI, declare it in `AI_USE.md` and log at least three errors you caught in `AI_ERRORS.md`

C3: Racing the News at Daily Frequency (1/2) [Proposed]

Solution discussed in class

Data and assumptions

- Lecture 2 case study: [Tak, Schlamp, Osterrieder & Pele \(2026\)](#) find that futures trades at Eurex and CME react to scheduled macro news within microseconds. This project asks what the same question looks like with daily closes, and what daily data can and cannot reveal.
- Data: S&P 500 (GSPC.INDX, close) and Euro Stoxx 50 (STOXX50E.INDX, close), daily log returns 2018–2025; the dates of the scheduled FOMC statements 2019–2025 from the [Federal Reserve calendar](#). The statement is released at 2:00 p.m. Eastern Time, after the European close.

Tasks

- (a) Build the event list: keep only scheduled statement days, drop unscheduled meetings, and state the number of events.
- (b) For the S&P 500, regress r_t^2 on a dummy D_t equal to 1 on statement days, with Newey–West (HAC) standard errors [Newey & West \(1987\)](#); test whether the variance is higher on event days.

C3: Racing the News at Daily Frequency (2/2) [Proposed]

Solution discussed in class

Tasks

- (c) Compute the average AR_0 and $CAR[+1, +5]$ with a constant-mean model estimated on days $[-250, -11]$ (the index is itself the market, so the market model of Lecture 2 cannot be used) [MacKinlay \(1997\)](#); semi-strong efficiency predicts $E[CAR[+1, +5]] = 0$.
- (d) Repeat (b) for the Euro Stoxx 50 twice, with the event on day 0 and on day +1, and explain which alignment is correct.

Report and interpret

- Report a table with the number of events, the D_t coefficients with HAC t -statistics, the average AR_0 and $CAR[+1, +5]$ with their t -statistics, and one slide that states what daily data confirm and what only nanosecond data can show (the speed of the first reaction, the speed rent). Model: B5, Seminar 0 Ex. 7.

Common Pitfalls

- ▣ Using $\pm 1.96/\sqrt{T}$ bands or homoskedastic z with volatility clustering: too many rejections
- ▣ Reading the best of many horizons, markets or calendar effects without a multiple-testing correction
- ▣ Taking an uncorrected R/S slope above 0.5 as proof of long memory
- ▣ Counting overlapping rolling windows as independent evidence
- ▣ Equating statistical rejection with a profitable strategy: costs and risk are missing
- ▣ Filling holidays with the previous close: creates zero returns and fake runs

Take-Home Assignment (Team Project)

- ▣ Choose one market not covered in class (another index, an exchange rate or a crypto-asset)
- ▣ Run the full battery: robust $\hat{\rho}_k$, $\tilde{Q}(10)$, runs, $z^*(q)$, Chow–Denning, Lo's V , DFA with a wild-bootstrap band
- ▣ Add a rolling analysis with a proper null band and one calendar effect with HAC errors and Holm correction
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: evidence, interpretation, limitations
- ▣ Log every AI error you catch, in this seminar or in the project, in `AI_ERRORS.md`

Further Study on Quantinar

Advanced courses on quantinar.com

-  Efficiency of cryptocurrency markets - A GMM-based analysis
-  Generalized Method of Moments

Code and data

- Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_02

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